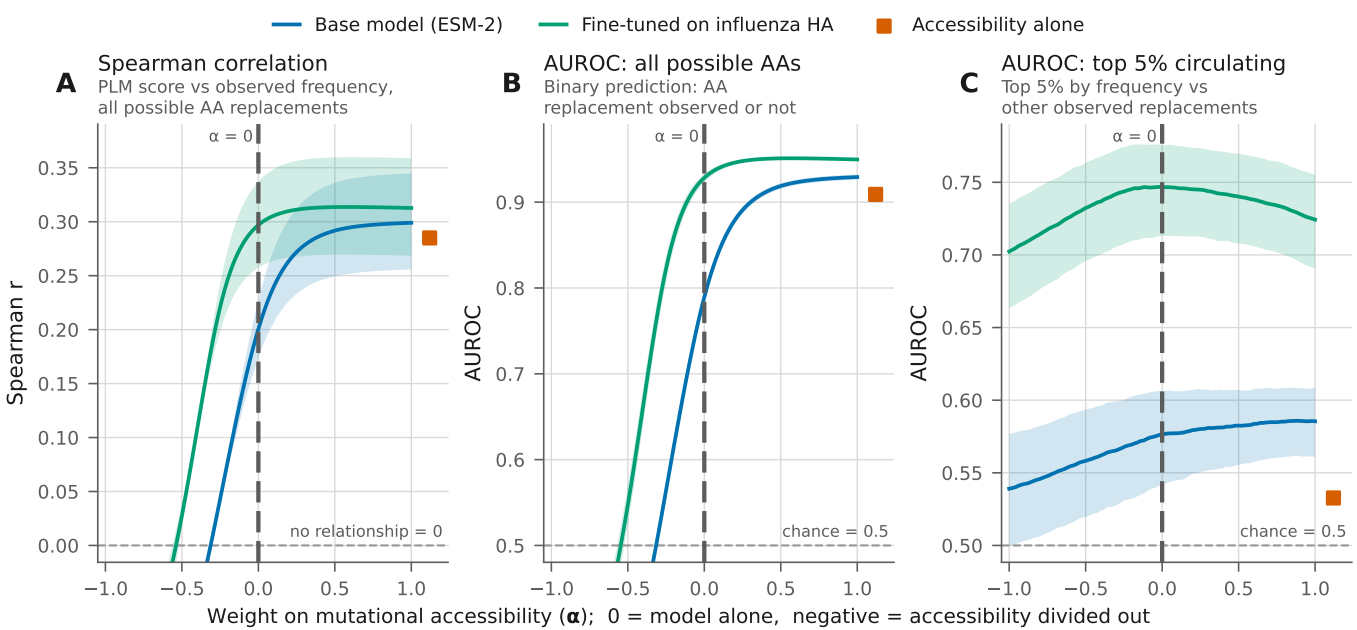
Base model (ESM-2)
Fine-tuned on influenza HA
Accessibility alone
A
Spearman correlation
PLM score vs observed frequency,
all possible AA replacements
Spearman r
α = 0
no relationship = 0
B
AUROC: all possible AAs
Binary prediction: AA
replacement observed or not
AUROC
α = 0
chance = 0.5
C
AUROC: top 5% circulating
Top 5% by frequency vs
other observed replacements
AUROC
α = 0
chance = 0.5
Weight on mutational accessibility (α); 0 = model alone, negative = accessibility divided out